



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTPP/OPERATION/2025-26/SYST_007/572002

Date:11/02/2026(18:05:53)

Subject:LMI on Operational Switching

The LMI for operational switching has been submitted for approval, in accordance with the requirements in the latest revision of OGN (OGN/OPS/SYST/007_Rev2).

No new revisions as per CC-OS.

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On:11/02/2026(18:05:53) **Ref:**572002

Note No:2

Checked and forwarded.

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On:13/02/2026(20:18:35) **Ref:**572002

Note No:3

LMI is in line with the OGN. Forwarded for further review.

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On:28/02/2026(17:03:34) **Ref:**572002

Note No:4

LMI is reviewed and found to be in line with the OGN.

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On:02/03/2026(11:16:14) **Ref:**572002

Note No:5

Noted the training aspects of the LMI attached, forwarded for review and approval, please.

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On:02/03/2026(13:09:27) **Ref:**572002

Note No:6

LMI is in line with the OGN. May please be approved.

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Note No:7

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On:03/03/2026(20:35:33) **Ref:**572002

Note No:8

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On:04/03/2026(09:23:15) **Ref:**572002

Note No:9

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On:04/03/2026(20:52:33) **Ref:**572002

Note No:10

The proposal pertains to approval for the LMI: **LMI/OGN/OPS/SYST/ 007Issue:01 Rev:05 Dt : Feb 2026**, for operational switching, has been submitted for approval, in line with the latest revised OGN (OGN/OPS/SYST/007_Rev2)

The previous Superseded Document is: **LMI/OGN/OPS/SYST/ 007Issue:01 Rev:04 Dt : Sept 2023**

The review frequency for the LMI is **2 yearly basis** or as and when needed, whichever is earlier.

No new revisions is there as per CC-OS.

LMI has been reviewed by concerned HODs and has been certified to be in line with the OGN, for switching operations of electrical equipments.

The Competent Authority for the Approval of the proposal is HOP

Submitted for kind approval please.

Submitted by:

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On:16/03/2026(21:40:34) **Ref:**572002

Note No:11

Approved & Submitted by:

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

OPERATIONAL SWITCHING

LMI/OGN/OPS/SYST/007
Issue:01 Rev:05 Dt: Feb 2026

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Operational Switching
2	LMI No.	LMI/OGN/OPS/SYST/007 Issue:01 Rev:05 Dt: Feb 2026
3	Reference Document	COS-OGN/OPS/SYST/007 Rev. No: 2 June 2023
4	Superseded Document	LMI/OGN/OPS/SYST/007 Issue: 1 Rev: 04 Dt: Sept 2023
5	Prepared By	Name: Meenal Ranta, Engineer (Operation)
7	Checked By	Name: Somnath Bhattacharya HOD (Operation)
8	Checked by	Name: Malay Kanti Jana HOD (EMD)
9	Checked by	Ananda Srinivas Vaka AGM (RLI)
10	Checked by	Navin Kishore AGM (EEMG)
11	Reviewed By	Name: Ahmad Kaunain Raza GM (O&M)
12	Approved By	Name: Ayaskanta Jena H O P (Simhadri)

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TITLE	Operational Switching
LMI No.	LMI/OGN/OPS/SYST/007 Issue:01 Rev:05 Dt : Feb 2026

1.0 Scope

Company practice and experience from human element failures / errors have ably demonstrated the need for a strict discipline to be applied to switching operations.

This LMI gives directives to operation persons with procedures covering switching discipline, training assessment and nomination. This LMI is intended to apply for all HT and LT switchgears including 400KV switchyard feeders to local and remote operational switching activities pertaining to Simhadri Thermal Power Station.

2.0 Superseded Documents

Operation Guidance Note "Operational Switching" LMI/OGN/OPS/SYST/007 Issue: 1 Rev: 04 Dt: Sept 2023.

3.0 Revision Notes

No new revision as per CC-OS.

4.0 Training, Assessment and Nomination

- 4.1** Practical training on the apparatus to be switched should be provided under appropriate supervision. Operational person shall be given a good understanding of the operational requirements, apparatus function, operational procedures and consequences of incorrect operation.

Resp.: SCE and EMD area Engineer

- 4.2** Training facility at EDC/RLI shall be utilized for persons who are doing switching operations of grid apparatus. Trainings shall be identified as appropriate to voltage levels of switching.

Resp.: SCE, HOD (O) and Head of RLI

- 4.3** Prior to nomination an assessment of the individual's competence should be carried out with a practical test and reference to training records.

Resp.: HOD (O) and Head of RLI

- 4.4** Refresher training needs shall be identified via individual ongoing performance assessment.

Resp.: SCE

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- 4.5** Training through OEM to be arranged by the electrical maintenance for switchgear switching operations. Regular training sessions through internal/external faculty may be arranged for electrical equipment switching.

Resp.: HOD (EMD)

- 4.6** Switching on electrical equipment's shall only be allowed by trained competent personnel. Certification may be done as per OGN-GEN-004 – "Guidelines for Qualification of SAPs, APs & CPs for PTW System".

Resp.: SCE

5.0 SWITCHING INSTRUCTIONS

- 5.1** Switching instructions and switching operations should be written down and retained in message book with details like SRLDC grid code, name of person giving instruction, person receiving instruction and time of receiving/transmitting instruction. These should cover switching operations controlled and carried out by one person as well as switching instructed by one person implemented by another.

Resp.: Switchyard Engineer, UCE, ASCE, SCE

- 5.2** When switching instructions are given by one person to another, there should be a discussion beforehand to ensure the objectives and effects of the operations are understood.

Resp.: Switchyard Engineer, UCE, ASCE, SCE

- 5.3** The person giving the instruction should obtain and record the name of the person who is going to carry out the switching. Instructions should include the following details: a) Name of the person giving the instruction. b) The location, identification and nomenclature of the apparatus. c) The switching operations to be carried out and their effect. d) Time and date of the instruction.

Resp.: Switchyard Engineer, UCE, ASCE, SCE

- 5.4** The person receiving the instruction should record it in accordance with procedures as follows - repeat the instruction back phrase by phrase as received and read back the instruction in full to ensure it has been received as intended.

Resp.: Switchyard Engineer, UCE, ASCE, SCE

- 5.5** Switching instructions must be given directly to the person who is to carry out the operation.

Resp.: Switchyard Engineer, UCE, ASCE, SCE

- 5.6** Where a person under training is to carry out the switching, that person must be under direct supervision of a trained and nominated person who is able to the switching operations.

Resp.: Switchyard Engineer, UCE, ASCE, SCE

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5.7 Any objections or queries at any stage of the instruction of the switching should be raised and resolved.

Resp.: Switchyard Engineer, UCE, ASCE, SCE

5.8 The person who is carrying out the switching instruction should observe the following:

Resp.: Switchyard Engineer, UCE, ASCE, SCE

5.8.1 Actions should be positively implemented without either rushing or causing undue delay but nothing should be taken for granted.

5.8.2 The person carrying out the operation must recheck and confirm that he is going to operate the correct apparatus.

5.8.3 Pause and re-check the proposed action.

5.8.4 Each single operation shall be carried out while filling the switching instruction sheet and relevant checks carried out using all available means to ensure it has been satisfactorily completed.

5.8.5 When a switch or apparatus shows any indication of distress, its condition should be reported immediately to the person who gave the instruction. All persons should be kept clear of such apparatus. Accessibility to that location should be blocked till the necessary help (CISF/competent person) arrived to the location.

5.8.6 Time of completion of the switching operation should be recorded in the switching instruction sheet.

5.8.7 The switching carried out shall be confirmed to the person who gave the switching instruction. The person who gave the instruction should record the switching carried out, who did the switching and the time in the message book.

5.9 Where switching is normally carried out under instruction from another person, but emergency conditions dictate switching without instruction, then as soon as possible the details of the switching carried out should be reported.

Resp.: Switchyard Engineer and Desk Engineer

5.10 The equipment to be switched on shall be confirmed for all clearances and pending PTWs.

Resp.: Switchyard Engineer and Desk Engineer

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5.11 Alarm annunciations and relay indications should be recorded in log book.

Resp.: Switchyard Engineer and Desk Engineer

6.0 Switching operation in switchyard: Annexure III

7.0 Switching operation in switchgear

7.1 There have been a many abnormal operations during process of isolation or normalization of the switchgears. As by design there are smaller workings clearances inside switchgears due to which generally any abnormality eventually leads to heavy arc flash affecting man/material. Hence utmost due care is required while working with any type of switchgear or electrical panels.

Resp.: UCE, SCE

7.2 Identification of correct unit & module where the switching operation to be done is very important. The instructions should clear state about the unit, switchgear/ module location.

Resp.: UCE, SCE

7.3 It is to be ensured by person doing switching operation that module have same identification as per instruction given. Any deviation to be discussed before performing operation.

Resp.: UCE, SCE

7.4 The procedure for switching operation shall be displayed in all switchgears in local language and English. Unit identification shall also be available while entering switchgear.

Resp.: AGM (O)

7.5 Person doing switching operation shall ensure use of proper PPEs like arc suits, non-contact voltage detector or test lamps to check live supply and proper tools like long handle fuse puller, screwdrivers, pliers etc. as per instructions given for the switching/isolation/normalization activity.

Resp.: UCE, SCE

7.6 Lockout/tagout to be properly maintained.

Resp.: UCE, SCE

7.7 Proper door closing of switchgear is to be ensured before any isolation and normalization activity on switchgears.

Resp.: UCE, SCE

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7.8 The CC-OS circular "IOM on Isolation and Normalization Procedure in LT switchgear" shall be followed.

Resp.: UCE, SCE

8.0 RECORDS

1. Switch yard Engineers Logbook
2. Switchyard Message book
3. Switching Instruction Format (Appendix1)

9.0 RESPONSIBILITY

HOD (O) is responsible for implementation of this LMI.

10.0 REFERENCE DOCUMENT

This LMI is based on OGN on the subject "Operational Switching" COS-OGN/OPS/SYST/007 Rev. No: 2 June 2023 and LMI/OGN/OPS/SYST/007 Issue: 1 Rev: 04 Dt: Sept 2023.

11.0 REVIEW

HOD (O) shall be responsible for reviewing this document on a 2-yearly basis or as necessary.

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*Annexure I***SWITCHING INSTRUCTION FORMAT**

DATE & TIME OF INSTRUCTION FOR ISOLATION/NORMALISATION	
INSTRUCTION SENDER NAME	
SWITCHYARD/SWITCHGEAR	
INSTRUCTION RECEIVER NAME	
EQUIPMENT IDENTIFICATION	
INSTRUCTIONS RECEIVED	1) 2) 3) 4)
ANY DEVIATIONS/ISSUES	1) 2)
OPERATION COMPLETE DATE & TIME	
COMPLETED OPERATION ACTIVITY - REPORTED PERSON	

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*Annexure II***Check list for Operation switching**

S. No	Description	Status (Yes/No)
1.	Written SOP is available & understood	
2	Pep talk carried out with all concerned. objective of job & effect on system is understood	
	The equipment to be switched shall be confirmed for all clearances and pending PTWs	
3	The person giving the instruction should obtain and record the name of the person who is going to carry out the switching	
4	Instruction contains. • name • The location, identification and nomenclature of the apparatus. • The time and date of the instruction. • The switching operations to be carried out and their effect. • The time and date of the instruction.	
5	Instructor is present & supervising the switching	
6	Detail checklist is available	
7	Joint protocol signed with EMD before start of work	
8	Joint protocol signed with EMD after completion of work	
9	A detailed SLD of switchyard is to be displayed in the switchyard control room for general awareness of operation and maintenance Engineers.	
10	Person doing switching is wearing PPE & Arc suits	

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Annexure III

Abnormal switching operation

There have been a few abnormal operations during process of isolation of the lines for maintenance work. These have led to spark over and subsequent bus faults, resulting in tripping of lines and units. In most of such incidences, little alertness and adherence of procedures / protocols would have averted the happening. Therefore, it is essential to have written down procedures and instructions with responsibility center, for isolation and normalization of EHV / HV systems.

Hence, following need to be strictly adhered while isolating / normalizing EHV / HV system:

1. Detailed checklist- bay wise is to be prepared, with responsibility center for isolation/ normalization.

Resp.: HOD (O)

2. Head of O&M is to identify authorized person for issuing and accepting permit to work for different voltage levels of the system. A register will be maintained at each unit control room and switchyard control room.

Resp.: Head of O&M

3. A joint protocol is to be prepared and is to be signed by operation and Electrical maintenance engineers before executing isolation/ normalization activity.

Resp.: SCE and AGM (EMD)

4. A detailed SLD of switchyard is to be displayed in the switchyard control room for general awareness of operation and maintenance Engineers.

Resp.: HOD (O) and AGM (EMD)

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5. The authorized person shall issue written isolation/ normalization slip with details of activity to be carried out to the operator who will carry it through out isolation/ normalization process.

Resp.: SCE

6. In case suitable authorized person is not available on a particular day, Head of O&M may nominate a selected person from electrical maintenance to oversee Isolation and Normalization.

Resp.: Head of O&M/SCE

7. General isolation and normalisation activities should be avoided during night and peak hours. However, in case of grid requirements/requests, same may be done with the permission of Head of O&M.

Resp.: SCE

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Annexure IV

CHECKLIST- EHV/SWITCHYARD (400 KV SYSTEM)

SLN O	CHECKS	RESPONSIBILITY	FREQUENCY	REMARKS
1	Discuss the sequence of operation with the competent persons before performing any switching activity.	Switchyard engineer	Before switching	
2	Get SRLDC, CPCC codes& Telephonic message from the authorized persons before doing any EHV switching activity.	Switchyard engineer	Before switching	
3	Feeder should be disconnected from the circuit before issuing any isolation/normalization slip	Switchyard engineer	After switching	
4	Check the particular feeder switchgear in local and confirm breaker is in off condition.	Switchyard engineer	After switching	
5	Recheck the details mentioned in isolation/normalization slip (feeder number, isolation/normalization details, sequence of isolation)	Switchyard engineer	After/before charging	
6	No switching operation to be done from local.	Switchyard engineer		
7	Check panel indications, permissive after Isolation/normalization of feeder.	Switchyard engineer	After/before charging	
8	If any close trip happens identify & get it rectified the problems before charging again.	Switchyard engineer & maintenance engineer	After switching	
9	Check for cross PTW before issuing normalization slips.	Switchyard engineer	After/before charging	
10	Cross check isolation/ normalization in the switchyard mimic along with the returned isolation/normalization slip. Ex: Earth switch position, isolator position	Switchyard engineer	After/before charging	
11	In case of any equipment being charged ensure men& material removed, Temporary earths are removed, and Panel is fully boxed up. In case of any remote equipment/feeder charging keep in touch with the remote end personnel while switching.	Switchyard engineer	Before switching	